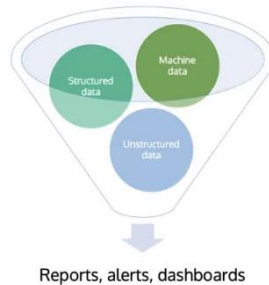


Splunk

Splunk is essentially a "Google for log files." It takes massive amounts of unstructured machine data (logs, metrics, clickstreams) and makes it searchable and visual in real-time.



Main Use Cases

1. **IT Operations:** Troubleshooting why a server went down by looking at logs.
2. **Security (SIEM):** Detecting a brute-force login attempt or a malware infection.
3. **Business Analytics:** Tracking how many customers clicked "Buy" in the last hour.

The Big Three Core Components Used in Splunk

To understand Splunk, you only need to know these three "actors" in the system:

- **Forwarder:** The "Collector." A lightweight agent installed on your servers that grabs log data and sends it to the Indexer.
- **Indexer:** The "Brain." It receives raw data, transforms it into searchable "events," and stores it on a disk.
- **Search Head:** The "Interface." This is the web GUI where you type queries. It sends your request to the Indexers and shows you the results.

The Data Pipeline: It defines how it works.

Data moves through four stages:

1. **Input:** Splunk consumes the raw data stream (files, ports, APIs).
2. **Parsing:** It breaks the data into individual events and identifies timestamps.
3. **Indexing:** It compresses the data and writes it to disk (creating "buckets").
4. **Searching:** You query the indexed data using SPL.



SPL: The Language of Splunk

SPL (Search Processing Language) is Splunk's version of SQL. It uses a "pipe" (|) system to pass results from one command to the next.

Basic Syntax Structure: search term | command | function

Common Commands:

- **search:** Filters data. (e.g., error OR failure)
- **stats:** Calculates statistics. (e.g., | stats count by host)
- **table:** Formats output into a table. (e.g., | table _time, user, ip)
- **fields:** Keeps or removes specific fields to speed up the search.
- **sort:** Orders your results.

Tip: Use wildcards (*) to match anything. status=4* will find 400, 404, and 500 errors.

Way of the "Search": When using Splunk, always follow the **Filter First** rule.

- **Bad Search:** * | search "error" (Tells Splunk to look at *everything* then find errors).
- **Good Search:** index=web_logs status=500 (Tells Splunk to go exactly to the right bucket and only grab errors).

Key Terminology

- **Source:** The specific file or stream the data came from (e.g., /var/log/httpd/access_log).
- **Sourcetype:** A category that tells Splunk how to format the data (e.g., linux_secure, cisco_syslog).
- **Index:** A logical bucket where data is stored. Separating data into indexes (e.g., index=security) makes searches much faster.
- **Knowledge Objects:** Saved searches, alerts, and dashboards that turn raw data into "intelligence."

Search Modes

When you run a query, you'll see a dropdown in the UI with three options. These dictate how much "heavy lifting" Splunk does while you wait for results:

Mode	Performance	Behavior
Fast Mode	Highest	Discovers only the fields required for your specific search. It disables "field discovery" to give you raw data back instantly.
Smart Mode	Balanced	The default. It switches between Fast and Verbose based on whether you are running a transformation command (like stats or chart).
Verbose Mode	Lowest	Discovers all possible fields in the data. Use this when you are exploring a new log file and don't know what fields exist yet.

Comparison between Fast, Smart and Verbose Mode

Attribute	Fast Mode	Smart Mode	Verbose
Speed	Fast	Balanced	Slow
Completeness	Poor	Balanced	Full
Field Discovery	Off	On	On
Event Available with stats?	No	On	Yes

Deployment Modes (How it's built)

This refers to the architecture or "topology" of your Splunk environment:

- **Standalone Mode:** Everything (Forwarder, Indexer, Search Head) lives on one single machine. Great for learning or small labs.
- **Distributed Mode:** The parts are separated across different servers. This allows you to scale to petabytes of data.
- **Clustered Mode:** Multiple Indexers or Search Heads work together. If one server dies, your data and searches stay online (High Availability).

Basic Ideas: The "Splunk Logic": To master Splunk, we need to grasp these three fundamental concepts:

A. Late-Binding Schema

Unlike a traditional database (SQL) where you must define columns *before* you upload data, Splunk uses **Schema-on-Fly**.

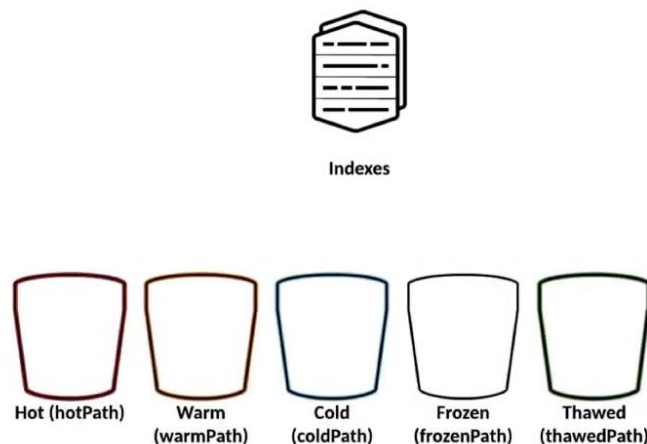
- We throw raw text into Splunk.
- Splunk only calculates what the "columns" are when you actually run a search.

Benefit: If your log format changes tomorrow, you don't have to rebuild the database.

B. The "Bucket" System

Splunk stores data in folders called **Buckets** based on time. As data gets older, it moves through stages:

1. **Hot:** Currently being written to (Searchable).
2. **Warm:** Recently written (Searchable).
3. **Cold:** Old data, often moved to slower storage (Searchable).
4. **Frozen:** Deleted or archived (Not searchable).



How Splunk Stores Data

C. Everything is an "Event"

Splunk doesn't care if it's a Windows log, a Tweet, or a temperature reading from a sensor. It treats every piece of data as an **Event**—a string of text tied to a specific **Timestamp**.